

p-Menthane hydroperoxide safety and handling precautions

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Compound of Interest

Compound Name: *p*-Menthane hydroperoxide

CAS No.: 80-47-7

Cat. No.: B1581428

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This technical guide provides comprehensive safety and handling information for **p-Menthane hydroperoxide** (PMHP), intended for researchers, scientists, and drug development professionals.

Overview of p-Menthane Hydroperoxide (PMHP)

p-Menthane hydroperoxide (CAS No: 80-47-7) is an organic peroxide used extensively as a radical initiator for emulsion polymerizations.^[1] It is instrumental in the production of polymers such as styrene-butadiene rubber (SBR), acrylonitrile butadiene styrene (ABS), and styrene-acrylonitrile (SAN), as well as acrylic and methacrylic polymers.^{[1][2][3]} Derived from turpentine, PMHP is also used as a cross-linking agent and in specialized chemical syntheses.^{[1][4]} It is typically supplied as a colorless to pale yellow liquid, often diluted with a solvent like isododecane.^{[1][2]}

Hazard Identification and Classification

PMHP is classified as a Type D or Type F organic peroxide, posing significant thermal and reactive hazards.^{[1][5]} It is a strong oxidizing agent that can undergo self-accelerating decomposition if heated or contaminated, potentially leading to fire or explosion.^{[1][6][7][8]} The

primary hazards include thermal instability, flammability, severe skin and eye damage, and harm if inhaled.[1][6][9][10]

Table 2.1: GHS Hazard Classification

Hazard Class	Category	Hazard Statement
Organic Peroxides	Type D / F	H242: Heating may cause a fire.[5][9][10][11]
Flammable Liquids	Category 3	H226: Flammable liquid and vapour.[5][9]
Skin Corrosion/Irritation	Sub-category 1A/1B	H314: Causes severe skin burns and eye damage.[5][9][10][11]
Serious Eye Damage	Category 1	H318: Causes serious eye damage.[5][9]
Acute Toxicity (Inhalation)	Category 4	H332: Harmful if inhaled.[10][11]
Specific Target Organ Toxicity (Repeated Exposure)	Category 2	H373: May cause damage to organs through prolonged or repeated exposure.[5][9]
Aspiration Hazard	Category 1	H304: May be fatal if swallowed and enters airways.[5][9]

Physical, Chemical, and Toxicological Properties

Understanding the properties of PMHP is critical for its safe handling.

Table 3.1: Physical and Chemical Properties

Property	Value	Source(s)
Molecular Formula	C ₁₀ H ₂₀ O ₂	[2][3]
Molecular Weight	172.26 g/mol	[1][2][3][10]
Appearance	Colorless to pale yellow liquid	[1][2][10]
Flash Point	53°C - 127°F (~53°C)	[1][4][7][10][11]
Boiling Point	~255.8°C at 760 mmHg	[4][11]
Density / Specific Gravity	0.91 - 0.96 g/cm ³	[1][4][7]
Solubility	Insoluble in water; Soluble in organic solvents (alcohols, esters, ethers, hydrocarbons)	[1][2][7][10]
Self-Accelerating Decomposition Temp. (SADT)	80°C	[3][12]
Theoretical Active Oxygen Content	~9.3%	[12]

Table 3.2: Toxicological Data

Metric	Value	Species	Notes
LD50 (Oral)	> 2000 mg/kg	Rat	[9]
LC50 (Inhalation)	Data not available	Rat	[9]
LD50 (Dermal)	Data not available	Rabbit	[9]

Safe Handling and Storage Protocols

Strict adherence to safety protocols is mandatory when working with PMHP.

Personal Protective Equipment (PPE)

- Eye/Face Protection: Wear chemical safety goggles and a full-face shield.[9][11]

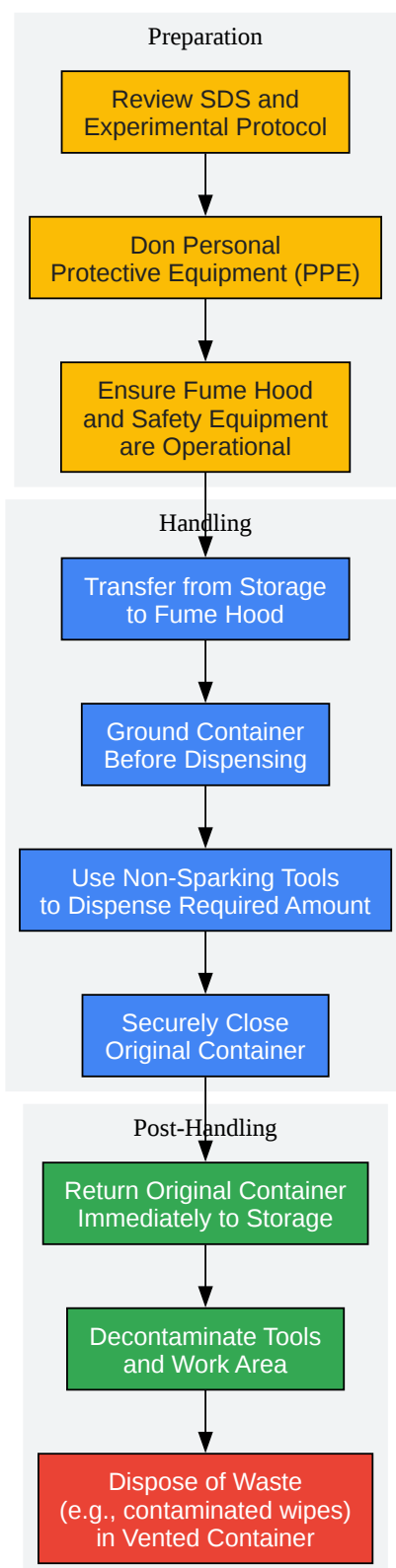
- Skin Protection: Use chemical-resistant gloves (e.g., neoprene or nitrile rubber) and a flame-retardant lab coat or apron.[9][11] Impervious clothing and shoes with conductive soles are recommended.[5][11]
- Respiratory Protection: Work in a well-ventilated area, preferably within a chemical fume hood.[1] If vapor concentrations are high or ventilation is inadequate, use a NIOSH-approved organic vapor respirator.[9]

Handling Procedures

- Keep away from heat, sparks, open flames, and other ignition sources.[1][13]
- Use only non-sparking tools and explosion-proof equipment.[5][9]
- Ground and bond containers and receiving equipment to prevent static discharge.[9][13]
- Avoid contact with skin, eyes, and clothing.[13]
- Never return unused material to the original container.[14]
- Minimize the quantity at the workplace to what is needed for a single shift or experiment.[15]

Storage Conditions

- Store in original, tightly closed containers in a cool, dry, well-ventilated area.[4][13][16]
- Protect from direct sunlight and store away from heat sources.[1][14][16]
- Maintain storage temperature below the recommended maximum (e.g., 25-30°C) and well below the SADT of 80°C.[3][12][14]
- Store separately from incompatible materials such as acids, bases, reducing agents, accelerators, combustible materials, and metals or rust.[6][9][15][16]
- Do not stack containers more than two high unless palletized.[15]



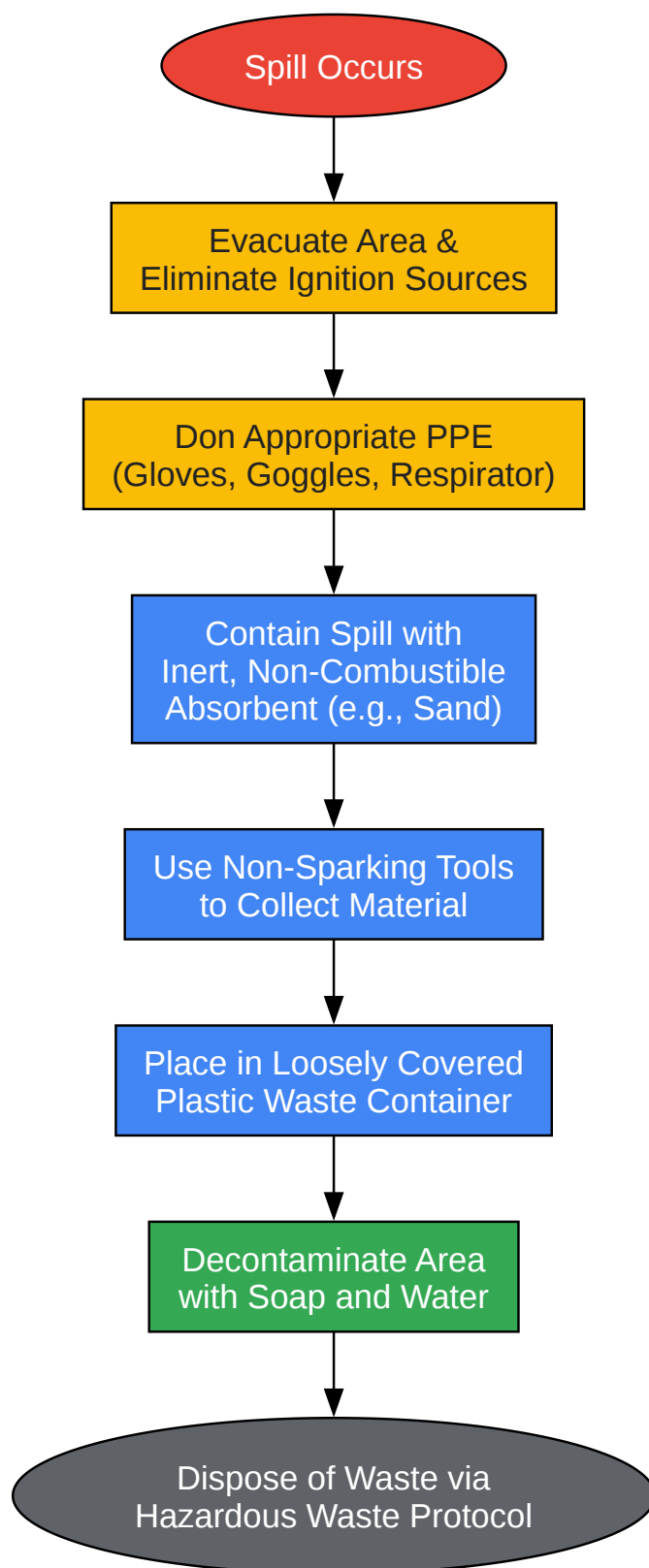
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Caption: Workflow for the safe handling of **p-Menthane hydroperoxide**.

Emergency Procedures

Spill Response

- Immediate Actions: ELIMINATE all ignition sources.^{[7][8]} Evacuate non-essential personnel and isolate the spill area (e.g., 50 meters for a liquid spill).^{[7][8]}
- Containment: For small spills, absorb with an inert, non-combustible, damp material like sand, vermiculite, or perlite.^{[7][10][16]} Do NOT use combustible absorbents like paper or cellulose-based materials.^{[7][8]}
- Cleanup: Using clean, non-sparking tools, place the absorbed material into a loosely covered plastic container for disposal.^{[7][10]} Do not seal the container tightly to allow for venting.^[17]
- Large Spills: Dike the spill to prevent spreading. Consider initial evacuation for at least 250 meters in all directions.^{[7][8]}



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Caption: Emergency procedure for a **p-Menthane hydroperoxide** spill.

Fire Fighting

- Suitable Extinguishing Media: For small fires, use water spray or fog, dry chemical, CO₂, or regular foam.[7][9][10] For large fires, flood the area with water from a distance.[7][11]
- Firefighter Precautions: Firefighters must wear self-contained breathing apparatus (SCBA) and full protective gear.[9] Cool containers exposed to fire with flooding quantities of water to prevent rupture and decomposition.[9][11] Fight fire from a maximum distance, as containers may explode when heated.[7][11]

First Aid Measures

- Inhalation: Move the person to fresh air immediately. If breathing is difficult, administer oxygen and seek immediate medical attention.[8][9]
- Skin Contact: Immediately remove all contaminated clothing.[9] Flood the affected skin with large amounts of water for at least 15 minutes, then wash gently with soap and water.[8][9] Seek immediate medical attention.[9]
- Eye Contact: Immediately flush eyes with copious amounts of water for at least 15-30 minutes, removing contact lenses if present.[8][9] Seek immediate medical attention.[8][9]
- Ingestion: Do NOT induce vomiting.[9][13] Rinse the mouth with water. If the person is conscious, have them drink plenty of water to dilute the substance.[9] Call a poison control center or doctor immediately.[9]

Experimental Protocol: Use as a Polymerization Initiator

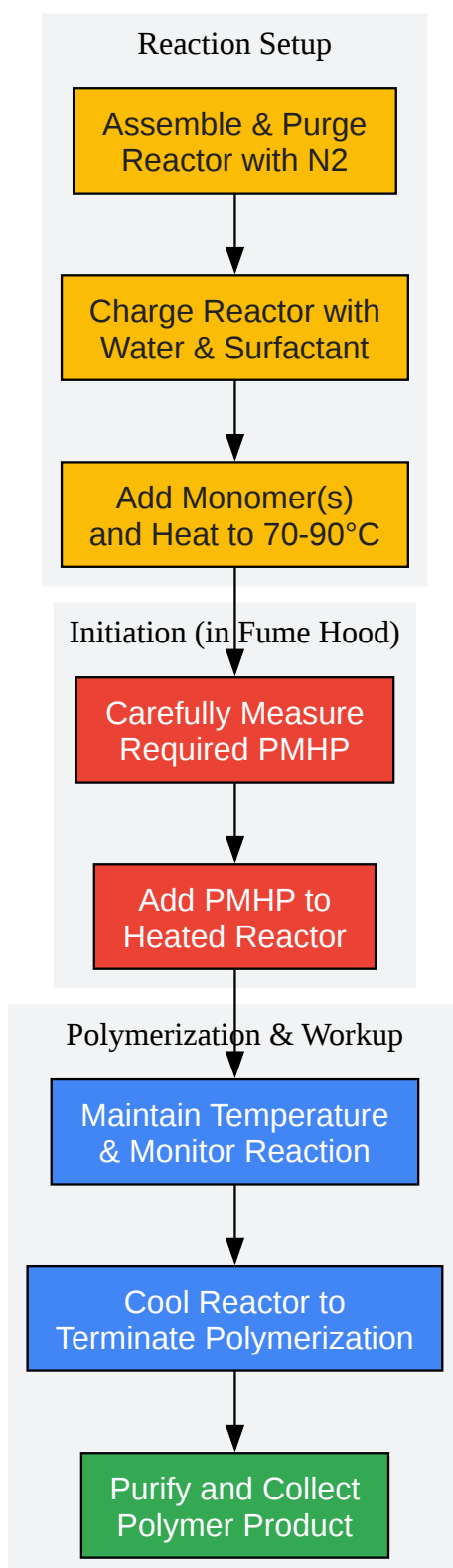
This section provides a generalized methodology for using PMHP as a radical initiator in an emulsion polymerization reaction. Note: This is a representative protocol. Researchers must adapt it based on specific monomer systems, desired polymer properties, and a thorough risk assessment.

Materials and Setup

- Reaction vessel equipped with a mechanical stirrer, reflux condenser, nitrogen inlet, and thermocouple.
- Monomer (e.g., styrene, acrylic acid).
- Surfactant solution (e.g., sodium dodecyl sulfate in deionized water).
- PMHP initiator solution (prepared in a suitable solvent if necessary).
- All manipulations of pure or concentrated PMHP must be done in a fume hood.

Methodology

- **Vessel Preparation:** The reaction vessel is charged with the surfactant solution and purged with nitrogen for at least 30 minutes to remove oxygen.
- **Monomer Addition:** The monomer is added to the vessel, and the mixture is stirred to form an emulsion. The temperature is raised to the target reaction temperature (typically 70-90°C).^[1]
- **Initiator Preparation (Safety Critical):** In a fume hood, the required amount of PMHP (e.g., 0.37–2.4 g per batch for SBR systems) is carefully measured.^[1] Avoid any contact with metal spatulas; use glass or compatible plastic pipettes.
- **Initiation:** The PMHP is added to the heated reaction mixture to initiate polymerization. The addition may be done in one portion or fed over time to control the reaction rate.
- **Polymerization:** The reaction is allowed to proceed at the set temperature for a predetermined time, with continuous stirring and nitrogen blanketing.
- **Termination and Workup:** Once the desired conversion is reached, the reaction is cooled, and any residual monomer is removed. The resulting polymer latex is then collected and purified.



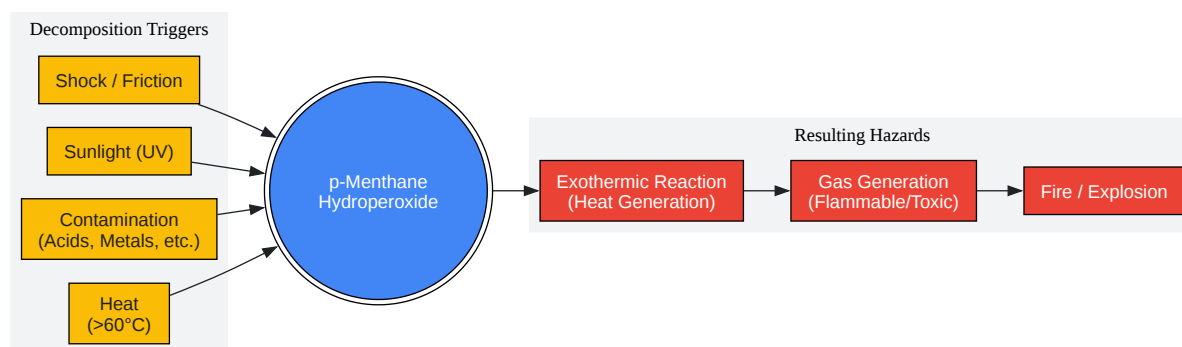
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Caption: Experimental workflow for PMHP-initiated emulsion polymerization.

Reactivity and Decomposition Hazards

The primary hazard of organic peroxides is their thermal instability.[6]

- Thermal Decomposition: PMHP will decompose exothermically if its Self-Accelerating Decomposition Temperature (SADT) of 80°C is reached.[6][12] This can create a runaway reaction, generating large volumes of flammable and toxic gases and potentially leading to a fire or explosion.[6][17] Explosive decomposition may occur at temperatures above 60°C.[9]
- Contamination: Decomposition can be triggered at temperatures much lower than the SADT by contamination with incompatible materials, including acids, bases, metal salts, and reducing agents.[6][9]
- Hazardous Decomposition Products: Thermal decomposition produces hazardous products including carbon monoxide, carbon dioxide, and other acid fumes.[9]



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Caption: Logical diagram of PMHP decomposition triggers and hazards.

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